

# **Neural Networks (2026)**

## **Course Information**

**Peter Lehoczký**

# Introduction

## Our Team (2026)

|                  |                                                                                 |
|------------------|---------------------------------------------------------------------------------|
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# Course Goals

- Inspire interest in neural networks
- Get in touch with latest technology



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- Inspire interest in neural networks
- Get in touch with latest technology
- **Have fun!**

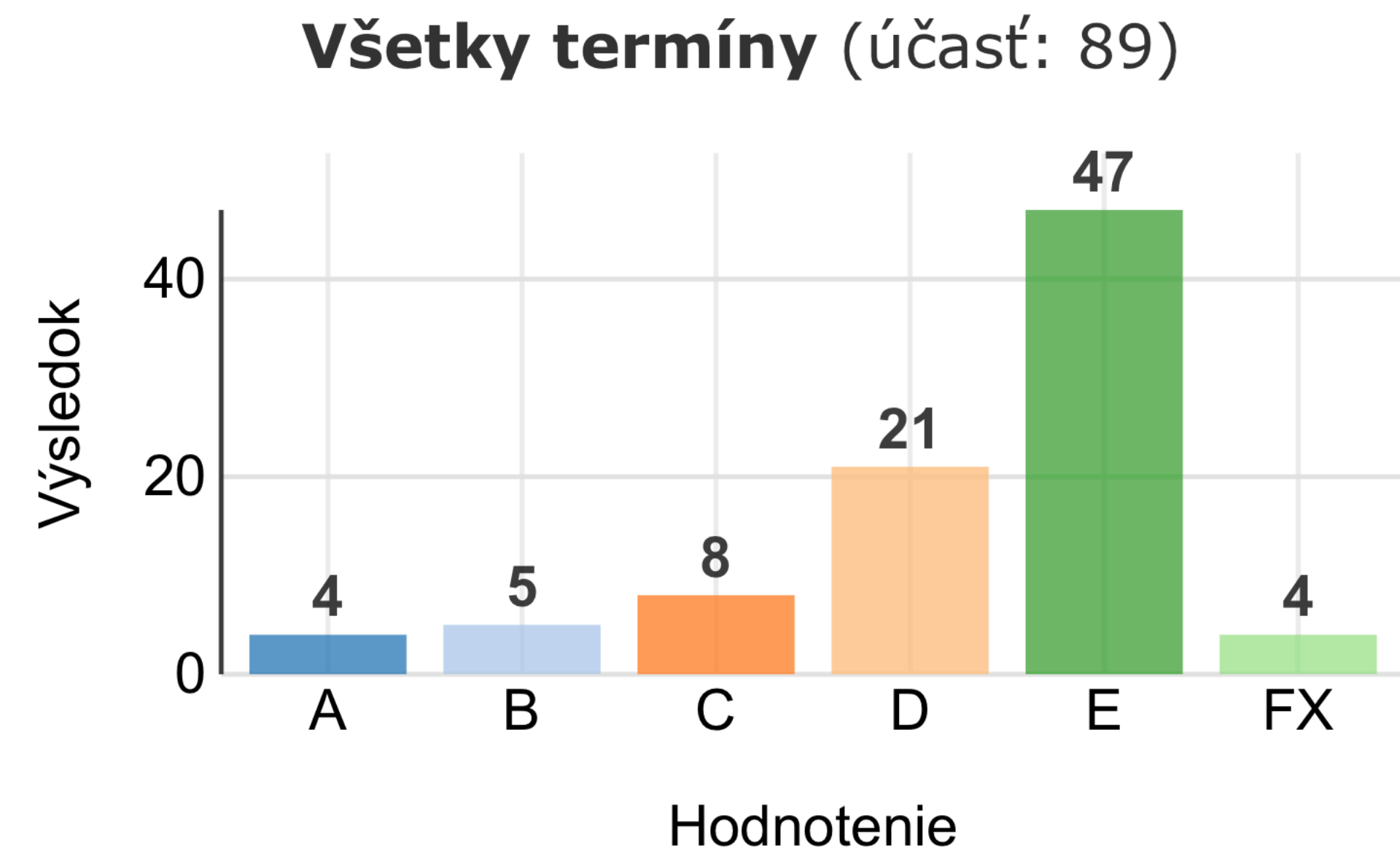
# Course Communication & Resources

## MS Teams: **FIIT NSIETE\_I - 2026**

- All students should already have access to the team - if not, please let me know
- Lecture slides will be available as PDFs
- Ask general questions in the channel - others may have the same ones
- Tip: check your notification settings so you don't miss updates

# Course Details

- **This is a difficult course**
- Bonus points\*
  - max. + 3 points during semester
  - \*note: bonus points do not count toward the minimum required points





# Rules

Min required

|            |                       |            |           |
|------------|-----------------------|------------|-----------|
| Semester   |                       | 60 points  |           |
|            | 3 Pre-lab Quizzes     | 3 points   |           |
|            | 3 Seminar Assignments | 7 points   | 2 points  |
|            | 3 Semestral Projects  | 35 points  | 15 points |
|            | Midterm Test          | 15 points  | 5 points  |
| Final Exam |                       | 40 points  | 10 points |
| Total      |                       | 100 points | 56 points |

# Seminar Assignments

- **Individual Assignments**
  - Week 2 - Forward Pass (2 points)
  - Week 3 - Backward Pass (2 points)
  - Week 4 - Optimizers (3 points)
- Submit into AIS



# 3 Semestral Projects

- **Teams Consisting of 2 Students**
- Week 5: start work; Week 6+: deliverables
- **Unique topic per team**
  - Exception: max 2 teams/topic with instructor approval
- **All 3 Semestral Projects Are Mandatory (5 points penalty per week of delay)**

# Semestral Projects

| Project Tasks                                                                                                                                                                                                                                        | Project 1 | Project 2 | Project 3 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|
| <ul style="list-style-type: none"><li>• Find, read and present SOTA paper/method</li><li>• Data analysis and processing</li><li>• Implementation</li><li>• Experimentation</li><li>• Evaluation (compare with SOTA)</li><li>• Presentation</li></ul> | 8 points  | 12 points | 15 points |

| Week | Seminar                                                    |  |  |
|------|------------------------------------------------------------|--|--|
| 1    | Introduction                                               |  |  |
| 2    | Assignment - Activation Functions, MLP, Forward Pass       |  |  |
| 3    | Assignment - Loss Functions, Derivatives, Back-propagation |  |  |
| 4    | Assignment - Optimizers                                    |  |  |
| 5    | Consultation 1 - Semestral Project 1                       |  |  |
| 6    | Presentation - Semestral Project 1                         |  |  |
| 7    | Consultation 1 - Semestral Project 2                       |  |  |
| 8    | Consultation 2 - Semestral Project 2                       |  |  |
| 9    | Presentation - Semestral Project 2                         |  |  |
| 10   | Consultation 1 - Semestral Project 3                       |  |  |
| 11   | Consultation 2 - Semestral Project 3                       |  |  |
| 12   | Presentation - Semestral Project 3                         |  |  |



| Week | Lecture Topic                                                           |
|------|-------------------------------------------------------------------------|
| 1    | Introduction, Neural Networks, Back-propagation of Errors               |
| 2    | Training, Regularization Techniques, Optimizers                         |
| 3    | Convolutional Neural Networks, Conv. Architectures                      |
| 4    | Application of CNNs (Object Localization, Detection, Segmentation, ...) |
| 5    | Recurrent Neural Networks                                               |
| 6    | NLP, Attention, Transformer                                             |
| 7    | Vision Transformers, Multimodal Models                                  |
| 8    | Midterm Test, Self-Supervised Pre-Training                              |
| 9    | Image Generative Models, Diffusion Models                               |
| 10   | Interpretability, Explainability of Neural Networks, Foundation Models  |
| 11   | Reinforcement Learning, Deep Reinforcement Learning                     |
| 12   | Special Guest Lecture (TBA)                                             |